

Andrew Mata

Stony Brook, NY | (631) 682-1493 | andrew.c.mata@gmail.com

[GitHub](#) | [LinkedIn](#) | [Google Scholar](#)

EDUCATION

Stony Brook University

Ph.D. in Computer Science

Stony Brook, NY

Aug. 2020 – Jun. 2026 (Expected)

Stony Brook University

M.S. in Computer Science

Stony Brook, NY

Aug. 2020 – Sept. 2023

Cornell University

B.A. in Computer Science and Mathematics

Ithaca, NY

Aug. 2016 – May 2020

TECHNICAL SKILLS

Languages: Python, Bash, MATLAB

ML / RL: PyTorch, Stable-Baselines3, Gymnasium, MuJoCo, NumPy, SciPy

Tools: Linux, Git, LaTeX

Research Areas: safe reinforcement learning, neural network verification, randomized smoothing, adversarial robustness

Mathematics: non-convex optimization, formal methods, optimal control, probability theory, linear algebra

EXPERIENCE

Stony Brook University, Reliable Systems Lab

Graduate Research Assistant, advised by Prof. Stanley Bak

Fall 2021 – Present

Stony Brook, NY

- Co-authored two peer-reviewed papers (ICCPS 2024, HSCC 2024) and presented a poster at ICCPS 2025 on adversarial attacks, falsification, and robustness certification of learning-enabled systems.
- Built Python/PyTorch frameworks for adversarial evaluation, MILP-based falsification, and robustness certification across Gymnasium and ARCH benchmark systems.

Air Force Research Laboratory (AFRL)

Research Intern, Autonomy Capability Team (ACT3)

Summer 2024

Dayton, OH

- Designed adversarial stress-tests for RL-based autonomous controllers in collaboration with ACT3 researchers.
- Implemented Python attack workflows integrating Zero-One Attack with ACT3's training infrastructure, enabling the team to benchmark new policies and surface failure modes.

SELECTED PROJECTS

Zero-One Attack: Adversarial Evaluation of RL Controllers | *Python, PyTorch, MuJoCo*

- Won **Best Artifact Evaluation Package at ICCPS 2024** for a reproducible Python/PyTorch framework featuring attack optimization, MuJoCo integration, baseline comparison, and trajectory visualization.
- Designed state-time perturbation attacks that degraded closed-loop neural controllers by 195% more than prior attacks across 4 complex MuJoCo control benchmarks.

Symbolic Gaussian Smoothing | *Python, PyTorch*

- Developed a deterministic robustness-certification method that propagates Gaussian moments symbolically through neural-network layers.
- Demonstrated lower computation times with comparable certification bounds compared to sampling-based baselines; presented as poster at ICCPS 2025.

PUBLICATIONS AND AWARDS

Best Artifact Evaluation Package Award, ICCPS 2024 (for Zero-One Attack).

Bak, S., **Mata, A.**, et al. "Zero-One Attack: Degrading Closed-Loop Neural Network Control Systems." *ICCPS*, 2024.

Bak, S., **Mata, A.**, et al. "Falsification using Reachability of Surrogate Koopman Models." *HSCC*, 2024.

Mata, A., Arjomandbigdeli, A., Bak, S. "Symbolic Gaussian Smoothing." *ICCPS 2025 (poster)*.